Experiment 7

Name: SOHAM TAYADE
Roll no. 52

Title

Use XSLT (eXtensible Stylesheet Language Transformations) to convert XML data into an HTML table for better presentation.Create an XML file (books.xml) containing book details (Title, Author, Price). Write an XSL file (books.xsl) that transforms the XML data into an HTML table. Link the XSL with XML and open it in a browser to see the formatted output

Theory:

XML (eXtensible Markup Language) is used to store and transport data in a structured and self-descriptive format. However, XML by itself does not define how data should be displayed in a web browser. To present XML data in a readable and user-friendly format, XSLT (eXtensible Stylesheet Language Transformations) is used.

XSLT is a language designed to transform XML documents into other formats such as HTML, XHTML, or plain text. It works by applying transformation rules defined in an XSL stylesheet to an XML document. These rules specify how elements in the XML file should be matched and displayed in the output document.

In this experiment, an XML file named books.xml is created to store book information such as title, author, and price. An XSL file named books.xsl is used to transform this XML data into an HTML table. The XSL file is linked to the XML document so that when the XML file is opened in a browser, the data is displayed in a formatted HTML table.

Algorithm / Procedure:

1. Start the experiment.
2. Create an XML file named books.xml.
3. Define book details such as title, author, and price using XML tags.
4. Create an XSL file named books.xsl.
5. Write XSLT rules to transform XML elements into an HTML table.
6. Link the XSL file to the XML file using the xml-stylesheet instruction.

7. Save both files in the same folder.
8. Open the XML file in a web browser.
9. Observe the formatted HTML table output.
10. Stop the experiment.

Program/Code:

1. **XML File (books.xml)**

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="books.xsl"?>

<books>
  <book>
    <title>Web Technology</title>
    <author>Deitel</author>
    <price>450</price>
  </book>

  <book>
    <title>XML and Web Services</title>
    <author>Harold</author>
    <price>520</price>
  </book>

  <book>
    <title>Learning PHP</title>
    <author>David Sklar</author>
```

```
    <price>399</price>
  </book>
</books>
```

XSL File (books.xsl)

```
<?xml version="1.0" encoding="UTF-8"?>

<xsl:stylesheet version="1.0"
    xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:template match="/">
<html>
<head>
  <title>Book Details</title>
  <style>
    table {
      border-collapse: collapse;
      width: 60%;
    }
    th, td {
      border: 1px solid black;
      padding: 8px;
      text-align: center;
    }
    th {
      background-color: #dddddd;
```

```
    }
  </style>
</head>

<body>
<h2 align="center">Book Information</h2>

<table align="center">
<tr>
  <th>Title</th>
  <th>Author</th>
  <th>Price</th>
</tr>

<xsl:for-each select="books/book">
<tr>
  <td><xsl:value-of select="title"/></td>
  <td><xsl:value-of select="author"/></td>
  <td><xsl:value-of select="price"/></td>
</tr>
</xsl:for-each>

</table>
</body>
</html>
```

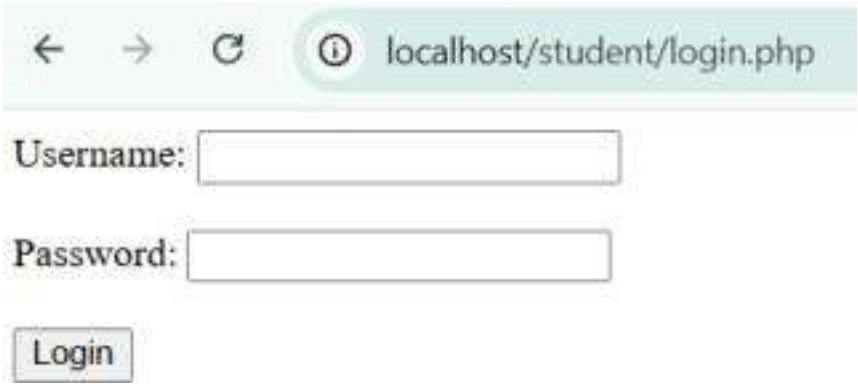
</xsl:template>

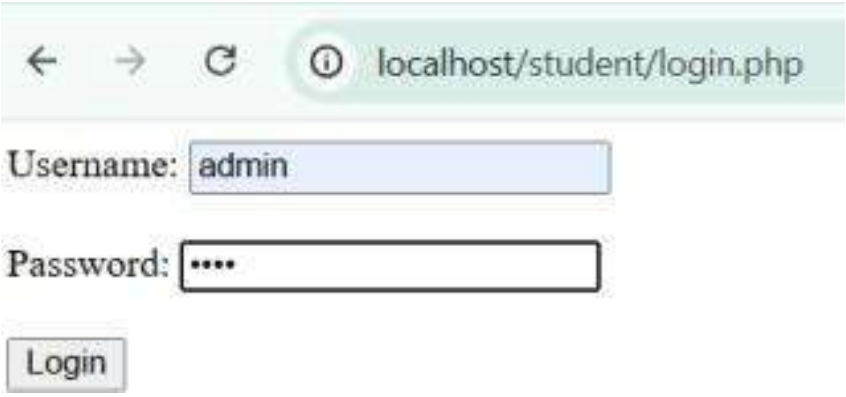
</xsl:stylesheet>

Scenario-Based Exercise:

A small college needs a simple web system to manage student details. The administrator should be able to add new students, view the student list, update student information, and delete records using a web page connected to a MySQL database through PHP. The system must validate user input before saving data, use sessions to allow only logged-in users to access the system, and use cookies to display a welcome message to the user.

Output:





localhost/student/dashboard.php

Welcome admin 🎉

Add Student | View Students | Logout

localhost/student/add.php

Student Added Successfully

Name:

Email:

Course:

Add Student

localhost/student/view.php

Student List

ID	Name	Email	Course	Action
3	John	john1234@gmail.com	BSc	Edit | Delete

localhost/student/edit.php?id=3

Edit Student

Name: John

Email: john1234@gmail.com

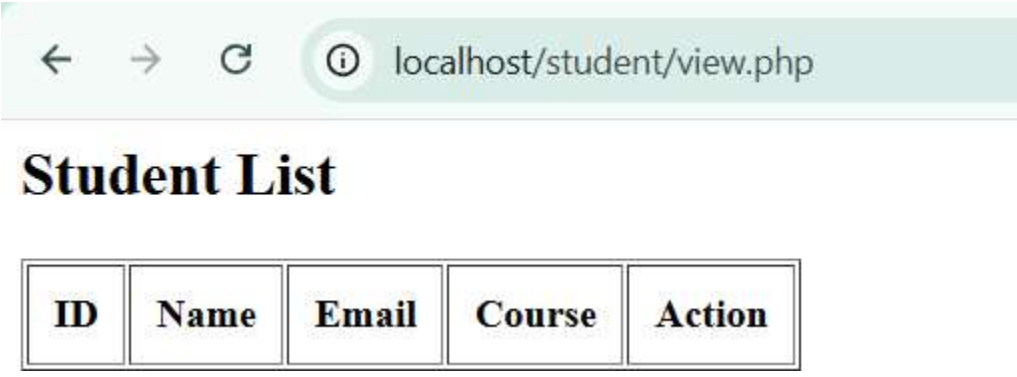
Course: BCA

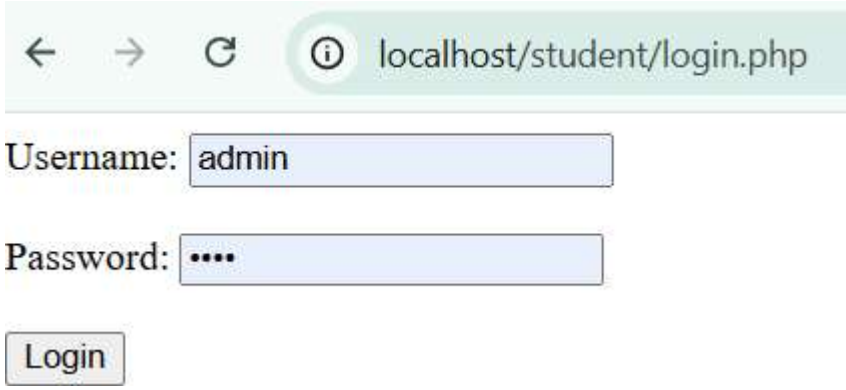
Update

localhost/student/view.php

Student List

ID	Name	Email	Course	Action
3	John	john1234@gmail.com	BCA	Edit | Delete





Conclusion

Thus, XML data was successfully transformed into an HTML table using XSLT. This experiment demonstrates how XSL stylesheets can be used to separate data storage from presentation and improve the readability of XML documents in web browsers.

Questions:

1. What is the primary purpose of XSLT in this experiment?

   ANS: C) To transform XML into HTML

   A) To store data
   B) To validate XML
   C) To transform XML into HTML
   D) To connect database

2. To extract the value of `<title>` element inside XSL, which tag is used?

Ans:B) `<xsl:value-of select="title"/>`

A) `<xsl:display>`
B) `<xsl:value-of select="title"/>`
C) `<xsl:get>`
D) `<xsl:print>`

3. What happens if XML tags are not properly closed?

ANS:C) XML becomes invalid

A) XSL ignores error

B) Browser displays partial output

C) XML becomes invalid

D) Only CSS fails